

SUSHMITA SENAPATI

MSc Neuroscience researcher with experience spanning experimental and computational neuroscience

sushmita.iitr16@gmail.com | sushmita.senapati@ovgu.de



EXPERIENCE

Master's Thesis Student in RG Synaptoscolopathies, OvGU, Magdeburg, Germany: May 2025 – Present

- Investigated hippocampal network dynamics underlying learning and memory using contextual fear conditioning and *in vitro* extracellular electrophysiology on a GABA-deficient transgenic mouse model
- Characterised sharp-wave ripple (SW-R) dynamics across ventral hippocampal CA3-CA1 circuits and quantified their differences between wild-type and knockout mice
- Developed a population firing-rate model of hippocampal CA1 to investigate how altered inhibitory activity influences SW-R incidence

HiWi in Comparative Neuroscience Unit, LIN Magdeburg, Germany: Apr 2024 – Feb 2025

- Developed an analysis pipeline for intracranial electrophysiological recordings from macaque primary auditory cortex to characterise neuronal responses to complex auditory stimuli
- Applied quantitative analysis to investigate population-level evoked neuronal responses

Volunteer at Ashadeep NGO, Guwahati, India: Jul 2022 – Dec 2022

- Took care of intellectually challenged individuals in a school, teaching them basic art, language, and maths

Bioinformatician at Corteva Agriscience Pvt. Ltd., Hyderabad, India: Jan 2020 – Jun 2022

- Employed Support Vector Machine (SVM) and an open-source tool, Sleuth, to perform differential gene expression (DGE) analysis
- Developed multiple synthetic promoter sequences using RNAseq and ATACseq data for enhancing maize crop yield

Bachelor's Thesis Student at IIT Roorkee, Uttarakhand, India: Jan 2019 – Mar 2019

- Analysed RNA-seq data from postmortem neuronal tissue to identify the abundance of LINE1 retrotransposon insertions in human patients with schizophrenia

Research Intern at OIST Graduate University, Okinawa, Japan: Nov 2018 – Jan 2019

- Perfected Zebrafish dissection and studied their eye lens equator formation in developing embryos using whole-cell antibody labelling

Research Intern at NCBS Bangalore, India: May 2018 – Jul 2018

- Identified targets for ribosome-binding protein Ataxin and differentially expressed genes in the brain of *Drosophila melanogaster* using R-code-based visualisations

Research Assistant at IIT Roorkee, Uttarakhand, India: Mar 2017 – Jul 2017

- Quantified gut microbiota composition of human siblings with differing body mass index to identify key microorganisms contributing to the difference using 2 pipelines - Qiime2 and Mothur

EDUCATION

M.Sc. in Integrative Neuroscience - Otto-von-Guericke-Universität (OvGU), Magdeburg, Saxony-Anhalt, Germany

Oct 2023 – Sept 2026

B.Tech. in Biotechnology - Indian Institute of Technology (IIT), Roorkee, Uttarakhand, India

Jul 2015 – May 2019

SKILLS

Computational tools: Python, MATLAB, R, bash, LINUX, git, LaTeX

Lab Tools and Techniques: *In vitro* (extracellular) electrophysiology, Contextual fear conditioning, Confocal microscopy, Immunohistochemistry

Additional Expertise: Spike Interface, XPPAUT, Dynamical Systems Theory, Linear Algebra, ODEs

POSTERS

Senapati, S., Pollali, E., Stork, O., Caliskan, G. Sharp Wave dynamics in response to population firing in GABA-deficient hippocampus: *In silico* approach. Presented at the 2025 Learning & Memory Conference, Magdeburg, Germany, September 2025.